K J SOMAIYAINSTITUTE OF TECHNOLOGY

Department of Information Technology

Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

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Division: IT-A
Batch: SA2
Roll Number: 34

Experiment No:2

Aim: ImplementationofQueueusingArrayforreal-worldapplication.

Program:

```
#include<stdio.h>
int q[50],i,n,choice,f=-1,r=-1,ele;

void dqueue(){
   if(f==-1){
      printf("\nQueue Underflow!!!(No active alerts)");
   }
   else{
      ele= q[f];
      printf("\nTemperature deleted: %d", ele);
      if(f==r){
         f=-1;
         r=-1;
      }
      else{
         f++;
      }
   }
}

 void enqueue(){
    if (r>=n-1){
      printf("\nQueue Overflow!!!");
   }
   else{
      printf("\nEnter the temperature to be inserted: ");
      scanf("%d",&ele);
      if (ele >= 45) {
         printf(">>> ALERT: Red Alert! Extreme Heatwave <<<\n");
      }
      else if (ele >= 43) {
         printf(">>> ALERT: Orange Alert! Severe Heatwave <<<\n");
      }
      else if (ele >= 40) {
         printf(">>> ALERT: Yellow Alert! Heatwave Warning <<<\n");
```

```
        }
        else {
            printf("\nTemperature is normal (< 40 deg). Clearing all active alerts...\n");
            while (f != -1) {
            dqueue();
            }
            printf("\nAll active alerts cleared successfully.\n");
            return;
        }
        r++;
        q[r]=ele;
        if(f==-1){
        f++;
        }
    }
}

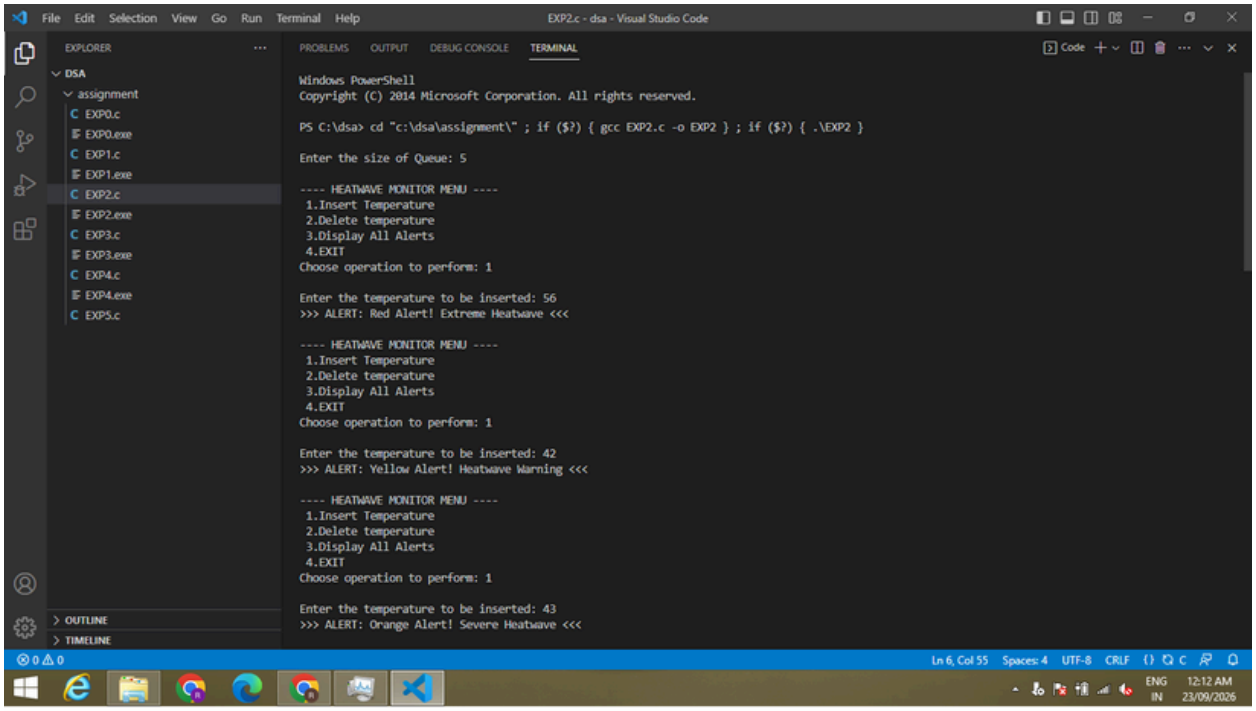
void display(){
    if(f==-1){
        printf("\nQueue Underflow!!!(No active alerts)");
    }
    else{
        printf("\nACTIVE ALERTS QUEUE:");
        for(i=f; i<=r; i++){
            printf("\n %d", q[i]);
        }
    }
}

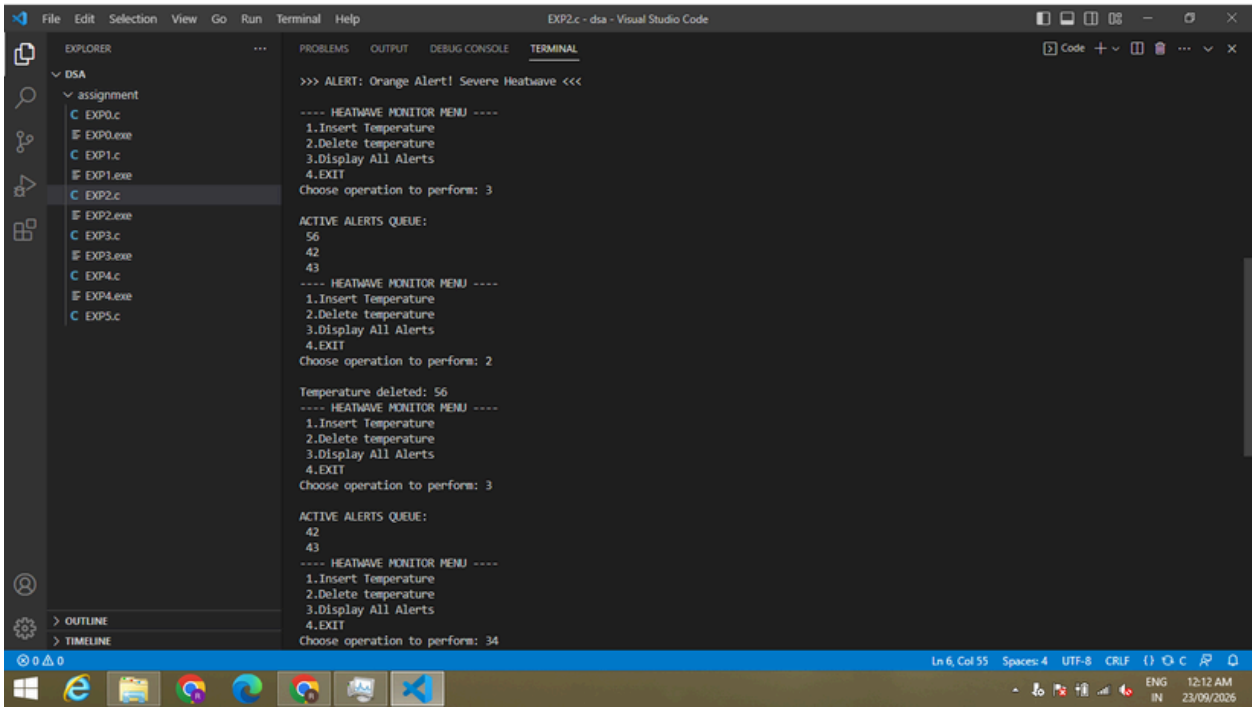
void main(){
    printf("\nEnter the size of Queue: ");
    scanf("%d", &n);
    do
    {
        printf("\n---- HEATWAVE MONITOR MENU ----\n 1.Insert Temperature \n 2.Delete
temperature \n 3.Display All Alerts \n 4.EXIT");
        printf("\nChoose operation to perform: ");
        scanf("%d", &choice);
        switch (choice)
        {
```

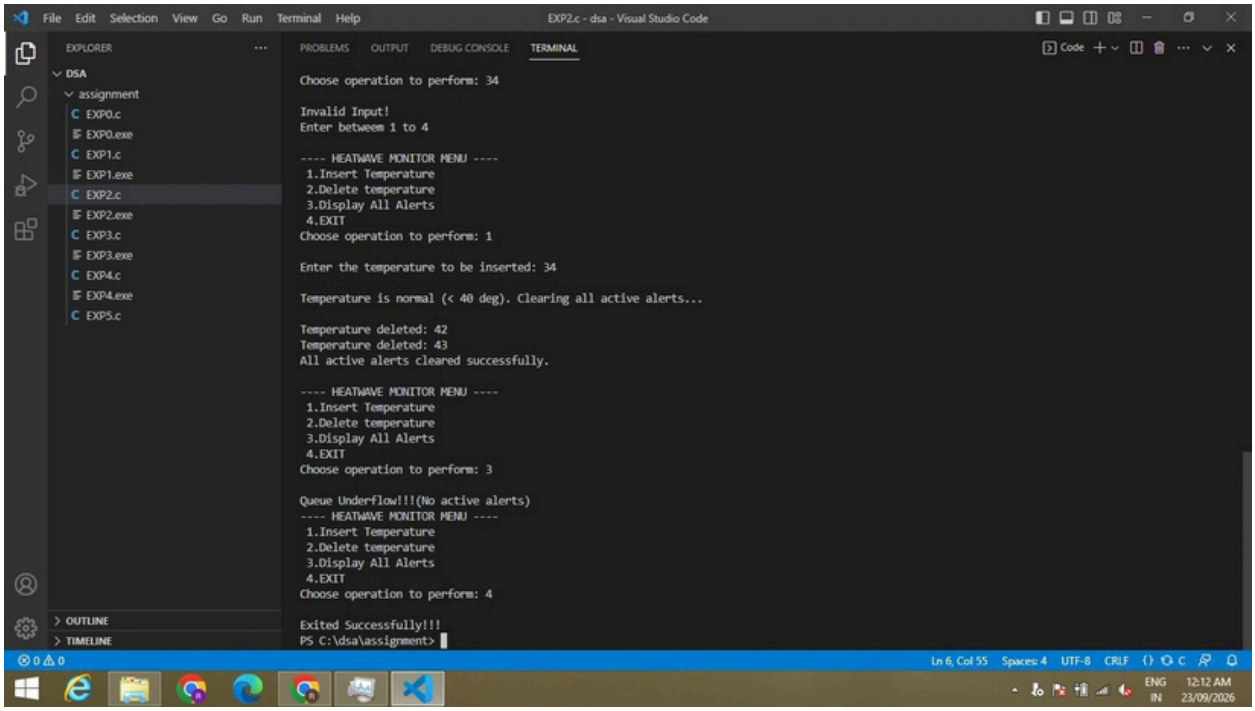
```
        case 1 :
            enqueue();
            break;
        case 2:
            dqueue();
            break;
        case 3:
            display();
            break;
        case 4:
            printf("\nExited Successfully!!!\n");
            break;
        default:
            printf("\nInvalid Input!\nEnter betweem 1 to 4\n");
            break;
        }
    }while (choice != 4);

}
```

Outcome:


File Edit Selection View Go Run Terminal Help
EXP2.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Windows PowerShell
Copyright (C) 2014 Microsoft Corporation. All rights reserved.
PS C:\dsa> cd "c:\dsa\assignment\" ; if ($?) { gcc EXP2.c -o EXP2 } ; if ($?) { .\EXP2 }
Enter the size of Queue: 5
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 1
Enter the temperature to be inserted: 56
>>> ALERT: Red Alert! Extreme Heatwave <<<
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 1
Enter the temperature to be inserted: 42
>>> ALERT: Yellow Alert! Heatwave Warning <<<
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 1
Enter the temperature to be inserted: 43
>>> ALERT: Orange Alert! Severe Heatwave <<<
Ln 6, Col 55 Spaces: 4 UTF-8 CRLF C
ENG IN 12:12 AM 23/09/2026


File Edit Selection View Go Run Terminal Help
EXP2.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
>>> ALERT: Orange Alert! Severe Heatwave <<<
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 3
ACTIVE ALERTS QUEUE:
56
42
43
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 2
Temperature deleted: 56
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 3
ACTIVE ALERTS QUEUE:
42
43
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 34
Ln 6, Col 55 Spaces: 4 UTF-8 CRLF C
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File Edit Selection View Go Run Terminal Help
EXP2.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Choose operation to perform: 34
Invalid Input!
Enter betweem 1 to 4
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 1
Enter the temperature to be inserted: 34
Temperature is normal (< 40 deg). Clearing all active alerts...
Temperature deleted: 42
Temperature deleted: 43
All active alerts cleared successfully.
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 3
Queue Underflow!!!(No active alerts)
---- HEATWAVE MONITOR MENU ----
1.Insert Temperature
2.Delete temperature
3.Display All Alerts
4.EXIT
Choose operation to perform: 4
Exited Successfully!!!
PS C:\dsa\assignment>
Ln 6, Col 55 Spaces: 4 UTF-8 CRLF C
ENG IN
12:12 AM 23/09/2026